Measurement-Efficient Port-Reading Policies for Fluid Antenna Systems Under Spatially Correlated κ-μ Shadowed Fading

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***Abstract*—Fluid antenna systems can exploit spatial diversity by switching among many candidate ports, but exhaustive port reading is often impractical in compact terminals. This work studies how to choose a small set of measured ports before port selection under spatially correlated κ-μ shadowed fading. For a 100-port aperture, three nested port-reading policies are compared: uniform subsampling, threshold-aware per-port ranking, and greedy coverage. The policies are learned and evaluated with repeated design/evaluation splits, and performance is measured by observed-only outage across 14 fading regimes and budgets $N \in \{3, 4, 5, 6, 7\}$. The greedy policy is strongest at the tightest budget, achieving the lowest outage in 12 of 14 regimes at $N = 3$. For $N \geq 4$, uniform subsampling becomes the most reliable overall and yields the best mean outage across regimes. In contrast, the ranking policy never attains the best mean outage, which indicates that ports with high marginal success probability can still be redundant in correlated apertures. The results establish port reading as a first-order design choice for measurement-efficient fluid antenna operation.**

***Keywords*—Fluid antenna systems, port selection, port reading, outage probability, spatial correlation, κ-μ shadowed fading, Monte Carlo.**

I. Introduction

Fluid antenna system has emerged as a promising antenna technology for beyond-5G and 6G systems, particularly when size and hardware constraints limit the use of conventional multi-antenna arrays [1]. By electronically moving a single radiating element across multiple candidate positions, an FAS can exploit spatial diversity with only one RF chain, which makes it attractive for compact terminals and sensor-class devices [2, 3]. When the best port is selected, the resulting diversity gain can approach or even exceed that of classical combining schemes despite the small aperture size [4]. However, these gains usually rely on reading many ports, which increases overhead and becomes impractical when channel measurements must be repeated frequently.

To reduce this burden, prior studies have considered partial-observation port selection, in which only a subset of ports is measured, and the final decision is made from the observed data or from a reconstructed port profile. Existing approaches include ML-based port selection, deep-learning-based transceiver strategies, and generative methods that infer unobserved ports from sparse measurements [5-9]. In all such approaches, the final decision depends not only on the predictor, but also on which ports are read.

This work focuses on that port-reading-stage question. Under a fixed budget of N measurements, the observed port set determines which spatial information is available and therefore constrains any subsequent selector or reconstruction method. This issue is especially relevant in spatially correlated apertures, where ports that appear informative individually may be redundant when read together. To isolate the effect of the port-reading pattern itself, this work evaluates the observed-only outage, namely, the outage obtained when selection is restricted to the measured ports.

A spatially correlated κ-μ shadowed dataset with $M = 100$ ports is used to compare three nested port-reading policies under strict observation budgets. Policy A is a geometry-driven uniform baseline. Policy B is a threshold-aware ranking policy based on empirical exceedance frequency. Policy C is a greedy coverage policy that iteratively selects complementary ports to reduce observed-only outage on a design split. The comparison is carried out with repeated design/evaluation splits across 14 fading regimes and budgets $N \in \{3, 4, 5, 6, 7\}$. The results show that Policy C is most effective at the tightest budget, whereas Policy A is the most robust overall choice for $N \geq 4$, while Policy B never attains the best mean outage.

The remainder of this paper is organized as follows. Section II presents the dataset construction and system model. Section III describes the port-reading policies and the evaluation protocol. Section IV reports the numerical results and discussion. Finally, Section V concludes the paper.

II. Dataset and System Model

A. Signal model and partial observation

A downlink scenario is considered in which a single-antenna BS serves a terminal equipped with an FAS composed of M candidate ports distributed over an aperture of electrical length D, expressed in wavelengths. Let $u \in \mathbb{C}$ denote the transmitted symbol with average power $\mathbb{E}[|u|^2] = P_u$. The baseband sample collected at port $m \in \{1, \ldots, M\}$ is modeled as

$$r_m = g_m u + w_m, \tag{1}$$

where g_m is the complex channel coefficient associated with port m and $w_m \sim \mathcal{CN}(0, N_0)$ is circularly symmetric complex Gaussian noise. The instantaneous SNR at port m is therefore

$$\eta_m = \frac{P_u}{N_0} |g_m|^2, \tag{2}$$

and the full SNR profile is collected in $\boldsymbol{\eta} = [\eta_1, \ldots, \eta_M]^\mathsf{T}$.

To reduce measurement overhead, only a subset of ports is observed. Let $\mathcal{O}$ denote the index set of measured ports with $|\mathcal{O}| = N \ll M$, and define the observation vector $\boldsymbol{\eta}_\mathcal{O} = [\eta_{m_1}, \ldots, \eta_{m_N}]^\mathsf{T}$ for $\mathcal{O} = \{m_1, \ldots, m_N\}$. Under partial observation, any observed-only selector or profile-reconstruction method must operate from $\boldsymbol{\eta}_\mathcal{O}$ rather than from the full profile $\boldsymbol{\eta}$.

B. Spatially correlated κ-μ shadowed fading

The ports are assumed to experience spatial correlation due to sub-wavelength sampling along the aperture. For isotropic 2D scattering in the azimuthal plane, the correlation between ports m and n is modeled through a Jakes-type matrix $\mathbf{R} \in \mathbb{C}^{M \times M}$ with entries

$$[\mathbf{R}]_{m,n} = J_0\left(\frac{2\pi|m-n|}{D(M-1)}\right), \tag{3}$$

where $J_0(\cdot)$ denotes the zeroth-order Bessel function of the first kind and D is the normalized aperture length in wavelengths. Following [6], let the eigendecomposition be $\mathbf{R} = \mathbf{V}\boldsymbol{\Sigma}\mathbf{V}^\mathsf{H}$, and define the correlation-shaping matrix $\mathbf{B} = \mathbf{V}\boldsymbol{\Sigma}^{1/2}$.

To capture both multipath clustering and shadowing on dominant components, the port-wise fading is constructed by adapting the physical κ-μ shadowed model to a spatially correlated multiport vector process. In particular, the noncoherent cluster superposition in (4) follows the model's canonical definition and is rewritten here with an explicit port index and correlation shaping [10, eq. (7)]. A physically motivated construction expresses the received power at port m as a sum over μ clusters,

$$|g_m|^2 = \sum_{\ell=1}^{\mu} |q_{m,\ell} + p_m|^2, \tag{4}$$

where $q_{m,\ell}$ represents the scattered component of cluster ℓ at port m, and p_m is the dominant component shared across clusters at that port. Spatially correlated scattered vectors are generated as $\mathbf{q}_\ell = \sigma_q \mathbf{B}\mathbf{z}_\ell$, where $\mathbf{q}_\ell = [q_{1,\ell}, \ldots, q_{M,\ell}]^\mathsf{T}$, $\mathbf{z}_\ell \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_M)$, and σ_q^2 controls the average scattered power. The dominant component vector is modeled as $\mathbf{p} = \xi \boldsymbol{\upsilon}$, where $\boldsymbol{\upsilon} \in \mathbb{C}^M$ is a deterministic steering vector with identical entries υ_0 and ξ is a Nakagami-m random variable with $\mathbb{E}[\xi^2] = 1$ that accounts for shadowing. Since all ports are identically distributed, the ratio of dominant to scattered power is enforced by choosing $|\upsilon_0|^2 = \kappa\sigma_q^2$, equivalently $\kappa = |\upsilon_0|^2/\sigma_q^2$. The parameters κ, μ, and m retain their standard interpretations as the dominant-to-scattered power ratio, the number of clusters, and the shadowing severity, respectively [11].

C. Port-selection metric

With full observation, the ideal FAS selects the port that maximizes the instantaneous SNR,

$$\eta^\star = \max_{1 \le m \le M} \eta_m. \tag{5}$$

Given a threshold η_{th}, the OP is defined as

$$P_{\text{out}} = \Pr(\eta^\star < \eta_{\text{th}}), \tag{6}$$

and is approximated through Monte Carlo evaluation over independent channel realizations. Since a closed-form expression for P_{out} is not available, it is estimated by evaluating the empirical ratio between the number of outage events and the total number of independent channel realizations. Under partial observation, two selection rules are particularly relevant. The observed-only selector chooses the best port among the measured indices, $\eta^{\text{obs}} = \max_{m \in \mathcal{O}} \eta_m$, which yields $P_{\text{out}}^{\text{obs}} = \Pr(\eta^{\text{obs}} < \eta_{\text{th}})$ and isolates the effect of the port-reading policy. Alternatively, a reconstruction-aided selector first forms $\widehat{\boldsymbol{\eta}}$ from $\boldsymbol{\eta}_\mathcal{O}$ and then selects $\widehat{m} = \arg\max_m \widehat{\eta}_m$.

D. Dataset construction

A multi-regime dataset is generated to promote robustness across heterogeneous fading conditions. For each parameter triple (κ, μ, m) in a grid of 14 configurations spanning Rayleigh-, Nakagami-, Rician-, and κ-μ-like regimes, 10^6 independent realizations of $\boldsymbol{\eta}$ are produced using $M = 100$ ports and electrical length $D = 1$ [12]. For each realization, the observed index set $\mathcal{O}$ is generated according to a port-reading policy and an observation budget $N \in \{3, 4, 5, 6, 7\}$. The complete set of parameter combinations is summarized in Table I.

Table I: Fading model parameters used in the dataset. The numerical placeholders 0.001 and 50 denote limiting cases tending to 0 and ∞, respectively.

Dataset	κ	μ	m	**Channel distribution**
1	0.001	1	0.001	Rayleigh, severe shadowing
2	0.001	1	2	Rayleigh, moderate shadowing
3	0.001	1	50	Rayleigh
4	0.001	2	50	Nakagami
5	0.001	5	50	Nakagami
6	5	1	0.001	Rician, severe shadowing
7	5	1	2	Rician, moderate shadowing
8	5	1	50	Rician
9	5	2	0.001	κ-μ, severe shadowing
10	5	2	2	κ-μ, moderate shadowing
11	5	2	50	κ-μ
12	5	5	0.001	κ-μ, severe shadowing
13	5	5	2	κ-μ, moderate shadowing
14	5	5	50	κ-μ

III. Port-Reading Policies and Evaluation Framework

A. Generic M-port profile prediction from N measurements

Under partial observation, some transceiver strategies attempt to reconstruct the full vector $\boldsymbol{\eta} \in \mathbb{R}^M$ from the measured entries $\boldsymbol{\eta}_{\mathcal{O}} \in \mathbb{R}^N$ before final port selection. This stage can be expressed through a mapping

$$\widehat{\boldsymbol{\eta}} = f\left(\boldsymbol{\eta}_{\mathcal{O}}\right), \tag{7}$$

where f may represent any estimator, including parametric ML regressors or model-based interpolators. When supervised learning is adopted, f is commonly fit by minimizing a reconstruction loss over a training set, for instance, the mean squared error

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{M}\sum_{m=1}^{M}\left(\widehat{\eta}_m - \eta_m\right)^2. \tag{8}$$

After estimating the full profile, the predicted best port index is obtained as $\widehat{m} = \arg\max_m \widehat{\eta}_m$. Although the choice of f governs the achievable reconstruction fidelity, the port-reading pattern $\mathcal{O}$ defines which information is available at the input and therefore constrains the downstream decision. The experiments reported here isolate that upstream effect through observed-only outage, while this subsection provides the broader reconstruction context in which the same port-reading policies may later be used.

B. Nested port-reading policies

Port reading is defined as selecting an ordered sequence of ports to probe, which induces a nested family of index sets $\{\mathcal{O}(N)\}_{N=1}^{M}$ such that $\mathcal{O}(N) \subset \mathcal{O}(N+1)$. Nested policies are particularly useful for embedded operation, as they enable graceful degradation when the observation budget changes, and they allow a single learned order to serve multiple budgets. The study compares three policies that produce a full M-port order.

Policy A is a geometry-driven baseline that uniformly subsamples the aperture. For a budget of N, the selected indices are evenly spaced between ports 1 and M, which prioritizes spatial coverage and requires no data-driven tuning.

Policy B is a threshold-aware ranking policy learned from a design subset. Let $\mathcal{D}_{\mathrm{des}}$ denote the set of design samples and define the exceedance indicator $b_{i,m} = \mathbf{1}\{\eta_m^{(i)} > \eta_{\mathrm{th}}\}$. Policy B scores each port by its empirical exceedance frequency,

$$\widehat{p}_m = \frac{1}{|\mathcal{D}_{\mathrm{des}}|}\sum_{i\in\mathcal{D}_{\mathrm{des}}} b_{i,m}, \tag{9}$$

and ranks ports in decreasing order of $\widehat{p}_m$. Deterministic ties are resolved by the design-set mean SNR at each port, and then by the port index. This policy favors ports that are frequently above threshold when considered individually.

Policy C is a greedy coverage policy that explicitly accounts for set-level redundancy. Starting from an empty set, the policy selects one port per step to maximally reduce observed-only outage on the design split. At step k, let $\mathcal{S}_{k-1}$ denote the already selected ports and let $\mathcal{U}_{k-1}$ denote the design samples that remain uncovered, meaning that all currently selected ports fail the threshold. The next port is selected as

$$m^\star = \arg\max_{m\notin\mathcal{S}_{k-1}} \sum_{i\in\mathcal{U}_{k-1}} b_{i,m}, \tag{10}$$

which maximizes the number of newly covered samples. If all design samples become covered before the target budget is reached, the remaining positions are filled using the Policy B ranking among the unselected ports to keep the ordering nested and deterministic.

C. Per-regime Monte Carlo evaluation protocol

The port-reading policies are evaluated independently within each of the 14 fading regimes, using a repeated-split Monte Carlo protocol. In each repetition, the regime samples are randomly permuted with a deterministic seed and split into a design subset and an evaluation subset, using an $80/20$ fraction. Policies B and C are learned on the design subset and then evaluated on the disjoint evaluation subset. Policy A is deterministic, yet it is evaluated on the same evaluation samples to enable paired comparisons.

The primary metric is the observed-only outage probability induced by the read ports, defined for a given index set $\mathcal{O}$ as

$$\widehat{P}_{\mathrm{out}}^{\mathrm{obs}}(\mathcal{O}) = \frac{1}{|\mathcal{D}_{\mathrm{eval}}|}\sum_{i\in\mathcal{D}_{\mathrm{eval}}} \mathbf{1}\left\{\max_{m\in\mathcal{O}} \eta_m^{(i)} < \eta_{\mathrm{th}}\right\}. \tag{11}$$

Algorithm 1: Monte Carlo evaluation of port-reading policies

Require: Regime dataset $\mathcal{D} = \{\boldsymbol{\eta}^{(i)}\}_{i=1}^{S}$ with $\boldsymbol{\eta}^{(i)} \in \mathbb{R}^M$; threshold η_{th}; budgets $\mathcal{N}$; repetitions R; evaluation fraction ρ.

for *each regime* **do**
  for $r \leftarrow 1$ **to** R **do**
    randomly split $\mathcal{D}$ into design $\mathcal{D}_{\text{des}}$ and evaluation $\mathcal{D}_{\text{eval}}$ with fraction ρ
    compute exceedance indicators $b_{i,m} = \mathbf{1}\{\eta_m^{(i)} > \eta_{\text{th}}\}$ on $\mathcal{D}_{\text{des}}$
    learn Policy B ranking by sorting ports by $\widehat{p}_m$ with deterministic ties
    learn Policy C order via greedy coverage on $\mathcal{D}_{\text{des}}$, with Policy B tie-breaks
    for *each* $N \in \mathcal{N}$ **do**
      evaluate $\widehat{P}_{\text{out}}^{\text{obs}}\big(\mathcal{O}(N)\big)$ on $\mathcal{D}_{\text{eval}}$ for Policies A–C
      evaluate the ideal outage on $\mathcal{D}_{\text{eval}}$ using all M ports
  aggregate repetition outputs into per-budget means and variability summaries across repetitions

For reference, the ideal baseline uses all M ports in the maximization. Since (11) depends only on exceedance events, the evaluation can equivalently operate on a binary threshold matrix, which preserves the metric exactly while simplifying the greedy computations in Policy C. Uncertainty is characterized through repetition-to-repetition variability induced by the repeated design/evaluation splits, and results are summarized by per-budget mean outage values across the R repetitions within each regime. To compare Policies A and C more directly, the paired difference

$$\Delta_r(N) = \widehat{P}_{\text{out}}^{\text{obs}}(\text{C}; r, N) - \widehat{P}_{\text{out}}^{\text{obs}}(\text{A}; r, N) \tag{12}$$

is computed within each repetition r on the same evaluation subset. The full evaluation procedure is summarized in Algorithm 1.

IV. Results and Discussion

The Monte Carlo comparison follows the protocol described in Section III-C. Each of the 14 fading regimes is processed independently with $R = 100$ repetitions, an $80/20$ design/evaluation split, and budgets $N \in \{3, 4, 5, 6, 7\}$. The outage threshold is fixed to $\eta_{\text{th}} = 0.8$ in the linear domain. For each repetition and budget, the observed-only outage of Policies A–C is computed on the evaluation samples, together with the ideal baseline that assumes all $M = 100$ ports are observable.

Table II: Percentage of regimes won by each policy at each budget.

Policy	$N = 3$ **(%)**	$N = 4$ **(%)**	$N = 5$ **(%)**	$N = 6$ **(%)**	$N = 7$ **(%)**
A	14.3%	85.7%	92.9%	100.0%	92.9%
B	0.0%	0.0%	0.0%	0.0%	0.0%
C	85.7%	14.3%	7.1%	0.0%	7.1%

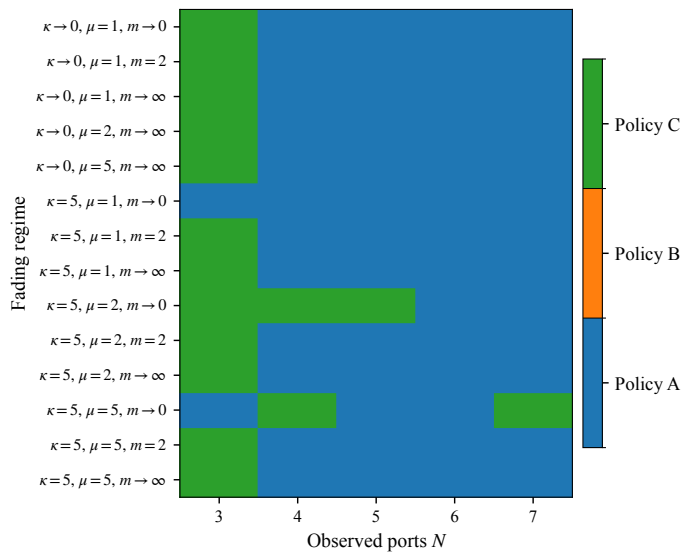

Figure 1: Best port-reading policy by fading regime and observation budget.

The regime-wise winners are summarized in Table II and visualized in Fig. 1. Policy C is strongest at the tightest budget and wins in 12 out of 14 regimes at $N = 3$. For $N \geq 4$, Policy A dominates and is the unique winner at $N = 6$, whereas Policy B never attains the lowest mean outage in the evaluated regimes.

Fig. 2 highlights the budget-dependent crossover between Policies A and C through the per-dataset paired mean difference in observed-only outage for $N = 3$ and $N = 4$. At $N = 3$, the differences are predominantly negative, which favors Policy C in most datasets. At $N = 4$, the sign becomes predominantly positive, which indicates that Policy A becomes preferable in most cases once one additional port can be read.

Fig. 3 illustrates the typical behavior in a representative regime. As expected, all policies improve with increasing N, because additional reads provide more opportunities to capture a port above threshold. Nonetheless, Policies A and C approach the ideal baseline much faster than Policy B. This reinforces the interpretation that set-level complementarity matters more than per-port ranking in spatially correlated apertures.

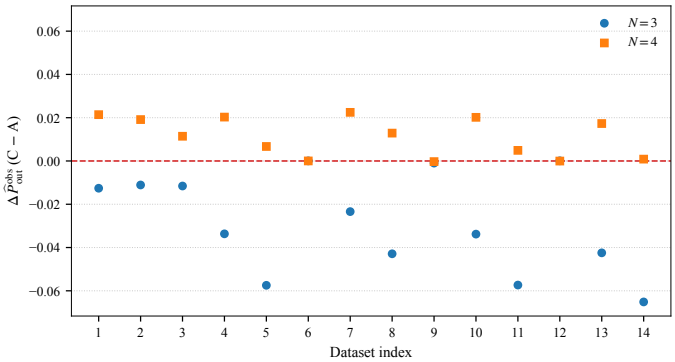

Figure 2: Forest plot of per-dataset paired mean differences in observed-only outage between Policies C and A for $N = 3$ and $N = 4$. Negative values favor Policy C.

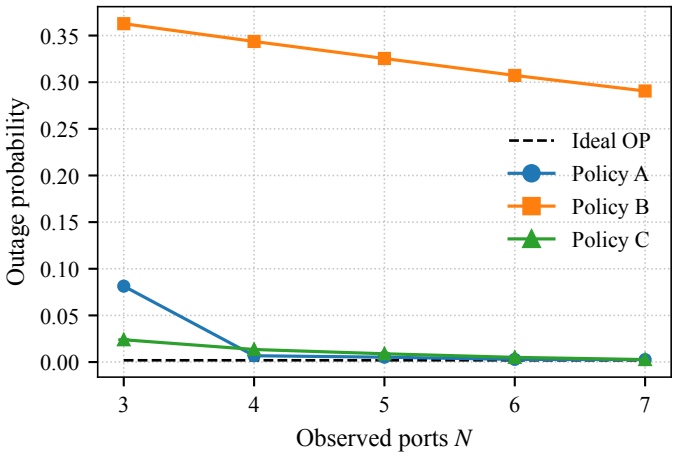

Figure 3: Observed-only outage versus number of observed ports in the $\kappa \to 0$, $\mu = 5$, $m \to \infty$ regime.

V. Conclusion

Port reading is a first-order design choice in FAS operation under partial observation, as it determines which spatial information is available to any observed-only selector or profile-reconstruction predictor. Under spatially correlated κ-μ shadowed fading with $M = 100$ ports, a repeated-split Monte Carlo evaluation across 14 regimes shows that a threshold-aware ranking strategy does not emerge as competitive in the reported comparisons and is therefore a poor default choice. The greedy coverage policy provides clear gains at the tightest budget, winning most regimes at $N = 3$, whereas uniform subsampling is the most robust default for $N \geq 4$ and dominates the majority of regime-budget settings. These findings indicate that complementarity-aware port reading can be valuable when measurements are extremely scarce, while geometry-driven uniform reading remains a strong baseline as the budget increases. Future work can extend this framework to reconstruction-aided selection, adaptive online port reading, and joint optimization of sensing policies under practical switching, latency, and hardware constraints.

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